Huawei Access Role QoS Congestion Event Root Cause Identification and Resolution SOP

Node: WFD-ACC-09 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_6445

Description

The QoS / Congestion Event (Microbursts, interface tail drops, QoS queue starvation) failure domain on Huawei Access Role equipment at the Whitefield POP site is triggered by the interaction of multiple underlying protocol and hardware mechanics. Specifically, this failure is precipitated by the interplay of the Access Control List (ACL) based policing and shaping mechanisms, the Quality of Service (QoS) policy maps, and the buffer allocation and management of the Physical Layer (PHY) and Data Link Layer (DLL) components within the Access node. When the policing and shaping mechanisms fail to match the incoming traffic with the pre-configured QoS policy, it can lead to microbursts of high-priority traffic being prematurely released onto the network, resulting in QoS queue starvation and interface tail drops. This is further exacerbated by the inefficient buffer allocation and management strategies of the PHY and DLL components, which can lead to a cascading effect of congestion events.

Pre-requisite Checks

- Verify the site location and confirm that the issue is isolated to the specified node, WFD-ACC-09, at the Whitefield POP.

- Confirm that the redundant link status is operational and functioning as expected.

- Ensure that the necessary remote access credentials have been obtained and the remote hands team is dispatched to the Whitefield POP site.

Potential Root Causes

Cause	Probability
Misconfigured QoS Policy Maps	High
Incorrectly Applied ACL Based Policing and Shaping Mechanisms	Medium
Inefficient Buffer Allocation and Management Strategies	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos policy-map, Expected Output: Policy map configuration, policing and shaping settings, and buffer allocation information, Description: This command is used to verify the configuration of the QoS policy maps, policing and shaping settings, and buffer allocation information on the Access node. It provides insight into the QoS policy configuration and the policing and shaping mechanisms in place, which can help identify potential misconfigurations or issues.
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```
Command: display acl based policing, Expected Output: Policing and shaping configuration, ACL based policing information, Description: This command is used to verify the configuration of the ACL based policing and shaping mechanisms on the Access node. It provides insight into the policing and shaping settings, ACL configuration, and buffer allocation information, which can help identify potential misconfigurations or issues.

Command: display buffer allocate, Expected Output: Buffer allocation information, PHY and DLL buffer allocation settings, Description: This command is used to verify the buffer allocation settings of the PHY and DLL components on the Access node. It provides insight into the buffer allocation strategies, which can help identify potential issues with inefficient buffer allocation and management.
```

Resolution Steps

Step 1: Step: Verify and Correct QoS Policy Map Configuration, Command: undo qos policy-map, Description: This command is used to correct any misconfigurations in the QoS policy maps, policing and shaping settings, and buffer allocation information. It ensures that the QoS policy maps are correctly configured and functioning as expected.

Step 2: Step: Correctly Apply ACL Based Policing and Shaping Mechanisms, Command: undo acl based policing, Description: This command is used to correct any misconfigurations in the ACL based policing and shaping mechanisms. It ensures that the policing and shaping settings are correctly applied and functioning as expected.

Step 3: Step: Optimize Buffer Allocation and Management Strategies, Command: buffer allocate optimize, Description: This command is used to optimize the buffer allocation strategies of the PHY and DLL components on the Access node. It ensures that the buffer allocation is efficient and functioning as expected.

Post-Check Verification

```
Command: display qos policy-map, Expected Output: Policy map configuration, policing and shaping settings, and buffer allocation information, Description: This command is used to verify that the QoS policy map configuration, policing and shaping settings, and buffer allocation information are correct and functioning as expected.

Command: display acl based policing, Expected Output: Policing and shaping configuration, ACL based policing information, Description: This command is used to verify that the ACL based policing and shaping mechanisms are correctly configured and functioning as expected.

Command: display buffer allocate, Expected Output: Buffer allocation information, PHY and DLL buffer allocation settings, Description: This command is used to verify that the buffer allocation strategies of the PHY and DLL components are efficient and functioning as expected.
```

Roll-back Procedures

- Step: Restore Original QoS Policy Map Configuration, Command: undo undo qos policy-map, Description: This command is used to restore the original QoS policy map configuration in case the fix fails.

- Step: Restore Original ACL Based Policing and Shaping Mechanisms, Command: undo undo acl based policing, Description: This command is used to restore the original ACL based policing and shaping mechanisms in case the fix fails.

- Step: Restore Original Buffer Allocation and Management Strategies, Command: undo buffer allocate optimize, Description: This command is used to restore the original buffer allocation strategies of the

PHY and DLL components in case the fix fails.